Lec 4. Sufficient and Complete Statistics

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Learning objectives

After this section, you should be able to ...

- under the optimal data reduction
- understand the definition of sufficient and complete statistics
- can find the sufficient and complete statistic if existed

4.1 Sufficient Statistics

Now we will talk about a special class of statistics called sufficient statistics(充分统计量). There are two ways to motivate sufficient statistics:

- **the data reduction viewpoint** where we would like to discard non-informative pieces of the dataset (for storage or other benefits), and
- **the risk reduction viewpoint** where we want to construct estimators that only depend on meaningful variation in the data. This will be studied in the next lecture.

The **goal in data reduction** roughly is to find ways to reduce the size of a dataset without throwing away important information. We need to fix ideas more concretely to make progress on this abstract question.

We focus on parametric models and suppose we observe samples

$$X_1, \ldots, X_n \sim f(x; \theta)$$

The typical statistical estimation problem is that we observe the samples as want to understand something about the unknown parameter θ. The goal of data reduction is to find statistics $T(X_1, \ldots, X_n)$ that contain all the information about the unknown parameter θ.

Definition

Sufficient Statistic: A statistic $T(X_1, \ldots, X_n)$ is said to be sufficient for the parameter θ if

$$f(x_1, \ldots, x_n \mid T(X_1, \ldots, X_n) = t; \theta)$$

does not depend on θ.

Example

Sufficiency of order statistics. Let $X = (X_1, \ldots, X_n)$ be iid according to an unknown continuous distribution F and let $T = \left(X_{(1)}, \ldots, X_{(n)}\right)$ be the ordered statitics, By the continuity assumptions, the X 's are distinct with probability 1 . Given T, the only possible values for X are the $n!$ vectors $\left(X_{(i_1)}, \cdots, X_{(i_n)}\right)$, and by symmetry, each of these has conditional probability $1/n!$ The conditional distribution is thus independent of F, and T is sufficient.

Example

Let $X_1 \ldots, X_n$ be a sample from $B(1, p)$, then $T = X_1 + \cdots + X_n$ is sufficient for p.

The Factorization Theorem

A much simpler characterization of sufficiency comes from what is called the **Neyman-Fisher factorization criterion (NFFC)**:

Theorem 1. *(因子分解定理) A statistic $T(X_1, \ldots, X_n)$ is sufficient for θ if and only if the joint pdf/pmf of $(X_1, \ldots, X_n)$ can be factored as*

$$p(x_1, \ldots, x_n; \theta) = h(x_1, \ldots, x_n) \times g(T(x_1, \ldots, x_n); \theta)$$

The factorization theorem is relatively straightforward to prove, at least in the discrete case.

Proof. **Factorization $\Longrightarrow$ sufficiency:**

$$
\begin{aligned}
p\left(x_1, \ldots, x_n \mid T=t ; \theta\right) &= \frac{p\left(x_1, \ldots, x_n, T=t ; \theta\right)}{p(T=t ; \theta)} \\
&= \frac{I\left(T\left(x_1, \ldots, x_n\right)=t\right) h\left(x_1, \ldots, x_n\right) \times g(t ; \theta)}{\sum_{x_1, \ldots, x_n: T\left(x_1, \ldots, x_n\right)=t} h\left(x_1, \ldots, x_n\right) \times g(t ; \theta)} \\
&= \frac{I\left(T\left(x_1, \ldots, x_n\right)=t\right) h\left(x_1, \ldots, x_n\right)}{\sum_{x_1, \ldots, x_n: T\left(x_1, \ldots, x_n\right)=t} h\left(x_1, \ldots, x_n\right)}
\end{aligned}
$$

which does not depend on θ.

Sufficiency $\Longrightarrow$ factorization: We simply define $g(t;\theta) = p(T = t;\theta)$, and $h\left(x_1, \ldots, x_n\right) = p\left(x_1, \ldots, x_n \mid T = t ; \theta\right)$, where by sufficiency we note that the latter function does not depend on θ. Now, it is straightforward to verify that factorization theorem holds. $\square$

Examples:

Corollary 1. *Under the assumptions of NFFC, a necessary and sufficient condition for a statistic T to be sufficient is that for any fixed θ and θ_0, the sample density ratio $p(\boldsymbol{x}, \theta)/p(\boldsymbol{x}, \theta_0)$ is a function only of $T(\boldsymbol{x})$.*

Proof. $\square$

Minimal sufficiency

The sufficient statistics are not unique. Furthermore, it seems, at least intuitively, that some sufficient statistics present much more reduction than others.

Definition

Minimal Sufficiency: A sufficient statistic T is minimal if for every sufficient statistic T' and for every $x, y \in \mathcal{X}$, $T(x) = T(y)$ whenever $T'(x) = T'(y)$. In other words, T is a function of T' (there exists f such that $T(x) = f(T'(x))$ for any $x \in \mathcal{X}$).

minimally sufficient statistic provides a greater reduction of the data than any other sufficient statistic.

Theorem 2. *Let $\mathcal{X}$ be the sample space and $\{p(x;\theta), \theta \in \Theta\}$ be a family of densities(pdf/pmf) on $\mathcal{X}$. Suppose that there exists a statistic T such that for every $x, y \in \mathcal{X}$:*

$$p(x;\theta) = C_{x,y} p(y;\theta) \quad \Longleftrightarrow \quad T(x) = T(y)$$

for every θ and some $C_{x,y} \in \mathbb{R}$ not depending on θ. Then T is a minimal sufficient statistic.

To prove this result, we first show that T is sufficient and then that it is minimal.

Proof. **T is sufficient:** Start with

$$T(\mathcal{X}) = \{t : t = T(x) \text{ for some } x \in \mathcal{X}\} = \text{range of } T.$$

For each $t \in T(\mathcal{X})$, consider the preimage $A_t = \{x : T(x) = t\}$ and select an arbitrary representative x_t from each A_t. Then, for any $y \in \mathcal{X}$ we have $y \in A_{T(y)}$ and $x_{T(y)} \in A_{T(y)}$. By the definition of A_t this implies that $T(y) = T\left(x_{T(y)}\right)$. From the assumption of the theorem,

$$\begin{aligned} p(y;\theta) &= C_{y,x_{T(y)}} p\left(x_{T(y)};\theta\right) \\ &= h(y) g_\theta(T(y)) \end{aligned}$$

which yields sufficiency of T by the NFFC.

*If $T(x) = T(y)$, such that $p(x;\theta)/p(y;\theta) = C_{x,y}$ does not depend on θ, then T is sufficient.

T is minimal: Consider another sufficient statistic T'. By the NFFC

$$p(x;\theta) = \tilde{g}_\theta\left(T'(x)\right)\tilde{h}(x).$$

Take any x, y such that $T'(x) = T'(y)$. Then

$$\begin{aligned} p(x;\theta) &= \tilde{g}_\theta\left(T'(x)\right)\tilde{h}(x) \\ &= \tilde{g}_\theta\left(T'(y)\right)\tilde{h}(y)\frac{\tilde{h}(x)}{\tilde{h}(y)} \\ &= p(y;\theta)C_{x,y} \end{aligned}$$

Hence, $T(x) = T(y)$ by the assumption of the theorem. So, $T'(x) = T'(y)$ implies $T(x) = T(y)$ for any sufficient statistic T' and any x, y. As a result, T is a minimal sufficient statistic.

*If $T(x)$ is sufficient, and $p(x;\theta)/p(y;\theta) = C_{x,y}$ not depending on θ implies $T(x) = T(y)$, then T is minimal sufficient.

□

Example

Suppose that $Y_1, \ldots, Y_n$ are i.i.d Poisson (θ).

$$p\left(y_1, \ldots, y_n; \theta\right) = \frac{e^{-n\theta}\theta^{\sum y_i}}{\prod y_i}, \quad \frac{p\left(y_1, \ldots, y_n; \theta\right)}{p\left(x_1, \ldots, x_n; \theta\right)} = \frac{\theta^{[\sum y_i - \sum x_i]}}{\prod y_i! / \prod x_i!}$$

which is independent of θ iff $\sum y_i = \sum x_i$. This implies that $T\left(X_1, \ldots, X_n\right) = \sum_{i=1}^{n} X_i$ is a minimal sufficient statistic for θ.

Remark 1. *What if the support of X, i.e., the set $\{x \in \mathcal{X} : p(x;\theta) > 0\}$, depends on θ? Then if $p(x;\theta) = C_{x,y}p(y;\theta), x$ and y must be supported by (exactly) the same θ 's. Otherwise there would be a θ dependence, which we assumed we did not have.*

Example

Let $X_1, \ldots, X_n \overset{iid}{\sim} U(0,\theta)$ and $T(X) = \max\{X_1, \ldots, X_n\}$. In that case for $x = (x_1, \ldots, x_n)$ such that $x_i > 0, i = 1, \ldots, n$ (in short, $x > 0$)

$$p(x;\theta) = \prod_{i=1}^{n} \frac{1}{\theta} I\left(x_i < \theta\right) = \frac{1}{\theta^n} I(T(x) < \theta).$$

If $T(x) = T(y)$ then $p(x;\theta) = 1 \times p(y;\theta)$. That scale ratio between the distributions does not depend on θ and so T is sufficient.
Conversely, if $x, y > 0$ are supported by the same θ 's, then $\{\theta$ supporting $x\} = (T(x), \infty) = (T(y), \infty) = \{\theta$ supporting $y\}$. Therefore $T(x) = T(y)$ and T is a minimal sufficient statistic.

Ancillarity(辅助性)

Although a minimal sufficient statistic provides in some sense an optimal degree of data compression, it is still possible for it to contain much "extra" or ancillary material that does not provide by itself any information about θ.

Definition

(Ancillarity). A statistic V is ancillary if its distribution does not depend on θ.

Example

Consider $X_1, \ldots, X_n \overset{iid}{\sim} \text{Cauchy}(\theta)$, whose distribution is given by

$$p(x;\theta) = \frac{1}{\pi}\frac{1}{1+(x-\theta)^2},$$

then $\left(X_{(1)}, \ldots, X_{(n)}\right)$ is minimal sufficient. While $R = X_{(n)} - X_{(1)}$ is ancillary statistic.

Proof. It is trivial to see the order statistics $T(X) = \left(X_{(1)}, \ldots, X_{(n)}\right)$ are sufficient, hence we only need to prove one direction: that if the ratio is constant as a function of θ, then $T(x) = T(y)$. That is, x must be a permutation of y.

Suppose $p(x \mid \theta) \propto p(y \mid \theta)$. Since this proportion holds for all θ, then divide each side by the case where $\theta = 0$ so that the constant of proportionality cancels and we get

$$\frac{p(x \mid \theta)}{p(x \mid 0)} = \frac{p(y \mid \theta)}{p(y \mid 0)}$$

Taking reciprocal gives us

$$\prod_i \frac{1 + (x_i - \theta)^2}{1 + x_i^2} = \prod_i \frac{1 + (y_i - \theta)^2}{1 + y_i^2}$$

Note that since these polynomials are equal, they must have the same roots. Also, each polynomial is of degree $2n$ and so they can have at most $2n$ roots. But it should be clear in this form that the LHS polynomial has complex roots

$$x_i \pm j$$

Since $(x_i - (x_i \pm j))^2 = (\pm j)^2 = -1$. While the RHS polynomial has roots

$$y_i \pm j$$

hence each side has $2n$ complex roots of that form, and since the polynomials share the same roots and cannot have more than $2n$ roots, it must be that x and y are permutations of one another, and so they have the same order statistics

$$T(y) = T(y)$$

thus $\left(X_{(1)}, \ldots, X_{(n)}\right)$ is minimal sufficient.

While the distribution of $R = X_{(n)} - X_{(1)} = \max(X_i - \theta) - \min(X_i - \theta)$ does not depend on θ, it is thus an ancillary statistic.

Ideally, the statistics that we make use of will include as little ancillary information as possible. In fact, we will demand even more than this. To do this we introduce a slightly weaker notion of ancillarity:

Definition

A statistic R is first-order ancillary if $E_\theta[R]$ does not depend on θ.

4.2 Completeness

Definition

(Completeness,完全性). A statistic T is complete for θ if for any θ, $E_\theta[h(T)] = c$ (constant) implies $h(T) = c$ with probability one. By subtracting the constant c this condition is equivalent to

$$E_\theta[h(T)] = 0 \quad \forall\, \theta \Rightarrow \quad h(T) = 0 \; wp.1$$

T is boundedly complete if the above holds for all bounded function h.

Note:

- the function h does not depend on any unknown parameters.
- In other words, a statistic T is complete for θ if no non-constant function of T is first-order ancillary. Completeness implies that two different functions of the statistics T cannot have the same expectation.
- Complete sufficient statistics yield optimal unbiased estimators. (This will be studied in the next lecture.)

Minimal sufficiency of complete sufficient statistic

Theorem 3 (Lehman and Schefle, 1950)**.** *If a minimal sufficient statistic exists, then any complete, and sufficient statistic is also a minimal sufficient statistic.*

Proof.

Note: A minimal sufficient statistic is not necessarily complete.

Example

A simple instance is $X \sim U(\theta, \theta+1)$ where $\theta \in \mathbb{R}$. It is not difficult to show X is a minimal sufficient statistic for θ. However,

$$E_\theta(\sin 2\pi X) = \int_\theta^{\theta+1} \sin(2\pi x)\mathrm{d}x = 0, \forall\theta$$

And $\sin 2\pi X$ is not identically zero almost everywhere, so that X is not a complete statistic.

Basu's Theorem

Theorem 4. *If T is a boundedly complete sufficient statistic for the family of distributions $\mathcal{P}_\theta$ then any ancillary statistic V is independent of T.*

Proof. By ancillarity, $p_C = P(V \in C)$ does not depend on θ for any C. Let $q_C(t) = P(V \in C \mid T = t)$. Then $E_\theta\left[q_C(T)\right] = p_C$, and so by completeness, $q_C(T) = p_C$ wp. 1. This proves the independence of V and T. $\square$

Remark 2. *For a statistic T, If a non-constant function of T, say $R(T)$ is ancillary, then T cannot be complete.*

Example

Let $X_1, \ldots, X_n$ be iid according to $U(\theta - 1/2, \theta + 1/2)$, $-\infty < \theta < \infty$. Here, $T = \left(X_{(1)}, X_{(n)}\right)$ is minimal sufficient. On the other hand, T is not complete since $X_{(n)} - X_{(1)}$ is ancillary.